

ERASTEEL TECHNICAL DATASHEET: ASP 2023

Application Family: Cutting tools | Standards: AISI M3:2, W.-Nr. 1.3395, HS 6-5-3

1. Grade Overview & Characteristics

ASP 2023 is a non-cobalt powder metallurgy high-speed steel characterized by excellent toughness, high dimensional stability during heat treatment, and balanced wear resistance. It is an optimal general-purpose grade across cutting tools and cold work.

2. Nominal Chemical Composition (wt. % range)

Element	Carbon (C)	Chromium (Cr)	Molybdenum (Mo)	Tungsten (W)	Vanadium (V)	Cobalt (Co)	Other
Range %	1.25 - 1.31	3.8 - 4.2	4.8 - 5.2	6.2 - 6.6	3.0 - 3.2	0.0	None

3. Mechanical Hardness & Property Profiles (Tempered state)

Metric	Value	Metric	Value (Rating 1-10)
Working Hardness Range	64.0 - 66.0 HRC	Machinability	6.0/10
Max Reachable Hardness	66.0 HRC	Wear Resistance	7.0/10
Annealed Delivery State	Max 280-350 HB	Toughness	7.5/10
Hot Hardness Retention	Secondary hardening	Hot Hardness	6.5/10
Heat Treatment Type	Vacuum / Salt bath	Grindability	6.0/10

4. Industrial Applications

Twist drills, taps, end mills, cold work punches and dies, fine blanking, powder compaction tools, rolls for cold rolling, shears.

5. Heat Treatment & Tempering Guidelines

Austenitizing at 1100-1180°C. Triple tempering at 560°C for 1 hour each. Max reachable working hardness 66 HRC.